Computed angle estimator with $\bar{A}'$

2026-10-24 states in (x, y, z): tracking error, the six entries of $\bar{A}$ and of $\bar{A}' = d\bar{A}/dh$

1 Motion gain matrix and its change along the surface normal

$$\bar{\mathbf{a}} = [\, \bar{A}_{xx} \;\; \bar{A}_{yy} \;\; \bar{A}_{zz} \;\; \bar{A}_{xy} \;\; \bar{A}_{xz} \;\; \bar{A}_{yz} \,]^{\mathsf{T}}$$

$$\bar{\mathbf{a}}' = [\, \bar{A}'_{xx} \;\; \bar{A}'_{yy} \;\; \bar{A}'_{zz} \;\; \bar{A}'_{xy} \;\; \bar{A}'_{xz} \;\; \bar{A}'_{yz} \,]^{\mathsf{T}}$$

$$\mathbf{M}[k] = \Delta\bar{\mathbf{p}}_d[k] + (1 - \lambda_c)\, \delta\bar{\mathbf{p}}_3[k]$$

$$\hat{\mathbf{M}}[k] = \Delta\bar{\mathbf{p}}_d[k] + (1 - \lambda_c)\, \delta\hat{\bar{\mathbf{p}}}_3[k]$$

2 True state of the process

$$\delta\bar{\mathbf{p}}_1[k+1] = \delta\bar{\mathbf{p}}_2[k]$$

$$\delta\bar{\mathbf{p}}_2[k+1] = \delta\bar{\mathbf{p}}_3[k]$$

$$\delta\bar{\mathbf{p}}_3[k+1] = \lambda_c\, \delta\bar{\mathbf{p}}_3[k] - \big(\bar{A} - \hat{\bar{A}}[k]\big)\, \bar{\mathbf{F}}_{dp}[k] - \boldsymbol{\varepsilon}_p[k]$$

$$\bar{\mathbf{a}}[k+1] = \bar{\mathbf{a}}[k] + \bar{\mathbf{a}}'[k]\, \mathbf{w}^{\mathsf{T}} \Big(\mathbf{M}[k] + \big(\bar{A} - \hat{\bar{A}}[k]\big)\, \bar{\mathbf{F}}_{dp}[k] + \boldsymbol{\varepsilon}_p[k] \Big)$$

$$\bar{\mathbf{a}}'[k+1] = \bar{\mathbf{a}}'[k] + \mathbf{w}_{a'}[k]$$

3 State vector and estimator

$$\mathbf{x}[k] = \big[\, \delta\bar{\mathbf{p}}_1^{\mathsf{T}} \;\; \delta\bar{\mathbf{p}}_2^{\mathsf{T}} \;\; \delta\bar{\mathbf{p}}_3^{\mathsf{T}} \;\; \bar{\mathbf{a}}^{\mathsf{T}} \;\; \bar{\mathbf{a}}'^{\mathsf{T}} \,\big]^{\mathsf{T}} \qquad (21 \times 1,\ d = 2)$$

$$\delta\hat{\bar{\mathbf{p}}}_1[k+1] = \delta\hat{\bar{\mathbf{p}}}_2[k] + \mathbf{L}_1[k]\, \mathbf{e}_y[k]$$

$$\delta\hat{\bar{\mathbf{p}}}_2[k+1] = \delta\hat{\bar{\mathbf{p}}}_3[k] + \mathbf{L}_2[k]\, \mathbf{e}_y[k]$$

$$\delta\hat{\bar{\mathbf{p}}}_3[k+1] = \lambda_c\, \delta\hat{\bar{\mathbf{p}}}_3[k] + \mathbf{L}_3[k]\, \mathbf{e}_y[k]$$

$$\hat{\bar{\mathbf{a}}}[k+1] = \hat{\bar{\mathbf{a}}}[k] + \hat{\bar{\mathbf{a}}}'[k]\, \hat{\mathbf{w}}^{\mathsf{T}}[k]\, \hat{\mathbf{M}}[k] + \mathbf{L}_4[k]\, \mathbf{e}_y[k]$$

$$\hat{\bar{\mathbf{a}}}'[k+1] = \hat{\bar{\mathbf{a}}}'[k] + \mathbf{L}_5[k]\, \mathbf{e}_y[k]$$

$\hat{\mathbf{w}}[k]$ from $\hat{\bar{A}}[k]$ by the plane formulas: $\tan 2\hat{\theta}_y = -2\hat{\bar{A}}_{xz}/(\hat{\bar{A}}_{xx} - \hat{\bar{A}}_{zz})$, $\tan 2\hat{\varphi}_{yz} = -2\hat{\bar{A}}_{yz}/(\hat{\bar{A}}_{yy} - \hat{\bar{A}}_{zz})$, $\hat{\theta}_x = -\arctan(\tan\hat{\varphi}_{yz} \cos\hat{\theta}_y)$

4 Error of the surface normal from the error of $\hat{\bar{A}}$

$\mathbf{e}_a = \bar{\mathbf{a}} - \hat{\bar{\mathbf{a}}} = [\, e_{xx}\ e_{yy}\ e_{zz}\ e_{xy}\ e_{xz}\ e_{yz}\,]^\mathsf{T}$; first order

$$
\begin{aligned}
e_{\theta_y} &= \frac{\left(\hat{\bar{A}}_{xx} - \hat{\bar{A}}_{zz}\right)\left(-2\, e_{xz}\right) + 2\,\hat{\bar{A}}_{xz}\left(e_{xx} - e_{zz}\right)}{2\left(\left(\hat{\bar{A}}_{xx} - \hat{\bar{A}}_{zz}\right)^2 + 4\,\hat{\bar{A}}_{xz}^2\right)} \\
e_{\varphi_{yz}} &= \frac{\left(\hat{\bar{A}}_{yy} - \hat{\bar{A}}_{zz}\right)\left(-2\, e_{yz}\right) + 2\,\hat{\bar{A}}_{yz}\left(e_{yy} - e_{zz}\right)}{2\left(\left(\hat{\bar{A}}_{yy} - \hat{\bar{A}}_{zz}\right)^2 + 4\,\hat{\bar{A}}_{yz}^2\right)} \\
e_{\theta_x} &= -\frac{\cos\hat{\theta}_y\left(1 + \tan^2\hat{\varphi}_{yz}\right)}{1 + \tan^2\hat{\varphi}_{yz}\cos^2\hat{\theta}_y}\, e_{\varphi_{yz}} + \frac{\tan\hat{\varphi}_{yz}\,\sin\hat{\theta}_y}{1 + \tan^2\hat{\varphi}_{yz}\cos^2\hat{\theta}_y}\, e_{\theta_y} \\
\mathbf{w} - \hat{\mathbf{w}} &= -\hat{\mathbf{v}}\, e_{\theta_x} + \cos\hat{\theta}_x\,\hat{\mathbf{u}}\, e_{\theta_y}
\end{aligned}
$$

5 Dynamics of estimation errors

F_x, F_y, F_z = components of $\bar{\mathbf{F}}_{dp}[k]$; $\mathbf{e}_{a'} = \bar{\mathbf{a}}' - \hat{\bar{\mathbf{a}}}'$

$$
\left(\bar{A} - \hat{\bar{A}}\right)\bar{\mathbf{F}}_{dp} = \begin{bmatrix} F_x & 0 & 0 & F_y & F_z & 0 \\ 0 & F_y & 0 & F_x & 0 & F_z \\ 0 & 0 & F_z & 0 & F_x & F_y \end{bmatrix} \mathbf{e}_a
$$

$$
\begin{aligned}
\mathbf{w}^\mathsf{T}\left(\mathbf{M} + \left(\bar{A} - \hat{\bar{A}}\right)\bar{\mathbf{F}}_{dp} + \boldsymbol{\varepsilon}_p\right) - \hat{\mathbf{w}}^\mathsf{T}\hat{\mathbf{M}} &= (1 - \lambda_c)\,\hat{\mathbf{w}}^\mathsf{T}\mathbf{e}_3 + \hat{\mathbf{w}}^\mathsf{T}\begin{bmatrix} F_x & 0 & 0 & F_y & F_z & 0 \\ 0 & F_y & 0 & F_x & 0 & F_z \\ 0 & 0 & F_z & 0 & F_x & F_y \end{bmatrix}\mathbf{e}_a \\
&\quad - \hat{\mathbf{v}}^\mathsf{T}\hat{\mathbf{M}}\, e_{\theta_x} + \cos\hat{\theta}_x\,\hat{\mathbf{u}}^\mathsf{T}\hat{\mathbf{M}}\, e_{\theta_y} + \hat{\mathbf{w}}^\mathsf{T}\boldsymbol{\varepsilon}_p
\end{aligned}
$$

$$
\begin{aligned}
\mathbf{e}_1[k+1] &= \mathbf{e}_2[k] \\
\mathbf{e}_2[k+1] &= \mathbf{e}_3[k] \\
\mathbf{e}_3[k+1] &= \lambda_c\,\mathbf{e}_3[k] - \begin{bmatrix} F_x & 0 & 0 & F_y & F_z & 0 \\ 0 & F_y & 0 & F_x & 0 & F_z \\ 0 & 0 & F_z & 0 & F_x & F_y \end{bmatrix}\mathbf{e}_a[k] - \boldsymbol{\varepsilon}_p[k] \\
\mathbf{e}_a[k+1] &= (1 - \lambda_c)\,\hat{\bar{\mathbf{a}}}'\,\hat{\mathbf{w}}^\mathsf{T}\mathbf{e}_3[k] + \mathbf{e}_a[k] + \hat{\bar{\mathbf{a}}}'\,\hat{\mathbf{w}}^\mathsf{T}\begin{bmatrix} F_x & 0 & 0 & F_y & F_z & 0 \\ 0 & F_y & 0 & F_x & 0 & F_z \\ 0 & 0 & F_z & 0 & F_x & F_y \end{bmatrix}\mathbf{e}_a[k] \\
&\quad - \hat{\bar{\mathbf{a}}}'\,\hat{\mathbf{v}}^\mathsf{T}\hat{\mathbf{M}}\, e_{\theta_x}[k] + \hat{\bar{\mathbf{a}}}'\,\cos\hat{\theta}_x\,\hat{\mathbf{u}}^\mathsf{T}\hat{\mathbf{M}}\, e_{\theta_y}[k] + \left(\hat{\mathbf{w}}^\mathsf{T}\hat{\mathbf{M}}\right)\mathbf{e}_{a'}[k] + \hat{\bar{\mathbf{a}}}'\,\hat{\mathbf{w}}^\mathsf{T}\boldsymbol{\varepsilon}_p[k] \\
\mathbf{e}_{a'}[k+1] &= \mathbf{e}_{a'}[k] + \mathbf{w}_{a'}[k]
\end{aligned}
$$

e_{θ_x}, e_{θ_y}: rows of Section 4, linear in $\mathbf{e}_a$

Error dynamics matrix

$$\mathbf{F}_e[k] = \begin{bmatrix} \mathbf{0} & \mathbf{I}_3 & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{I}_3 & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \lambda_c\,\mathbf{I}_3 & -\begin{bmatrix} F_x & 0 & 0 & F_y & F_z & 0 \\ 0 & F_y & 0 & F_x & 0 & F_z \\ 0 & 0 & F_z & 0 & F_x & F_y \end{bmatrix} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & (1-\lambda_c)\,\hat{\bar{\mathbf{a}}}'\,\hat{\mathbf{w}}^{\mathsf{T}} & \mathbf{I}_6 + \hat{\bar{\mathbf{a}}}'\,\hat{\mathbf{w}}^{\mathsf{T}} \begin{bmatrix} F_x & 0 & 0 & F_y & F_z & 0 \\ 0 & F_y & 0 & F_x & 0 & F_z \\ 0 & 0 & F_z & 0 & F_x & F_y \end{bmatrix} & \left(\hat{\mathbf{w}}^{\mathsf{T}}\hat{\mathbf{M}}\right)\mathbf{I}_6 \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{I}_6 \end{bmatrix}$$

$\hat{\mathbf{w}}$ taken as given in $\mathbf{F}_e$: the terms $-\hat{\bar{\mathbf{a}}}'\,\hat{\mathbf{v}}^{\mathsf{T}}\hat{\mathbf{M}}\,e_{\theta_x} + \hat{\bar{\mathbf{a}}}'\cos\hat{\theta}_x\,\hat{\mathbf{u}}^{\mathsf{T}}\hat{\mathbf{M}}\,e_{\theta_y}$ are dropped (about $0.0025\,\mathbf{e}_a$ next to $\mathbf{I}_6$; unbounded where $\hat{\bar{A}}_{xx} - \hat{\bar{A}}_{zz}$, $\hat{\bar{A}}_{xz}$ vanish)

6 Process outputs and output matrix

$$\mathbf{y}[k] = \begin{bmatrix} \delta\bar{\mathbf{p}}_m[k] \\ \bar{\mathbf{a}}_m[k] \end{bmatrix}, \qquad \hat{\mathbf{y}}[k] = \begin{bmatrix} \delta\hat{\bar{\mathbf{p}}}_1[k] \\ \hat{\bar{\mathbf{a}}}[k] \end{bmatrix}, \qquad \mathbf{H} = \begin{bmatrix} \mathbf{I}_3 & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{I}_6 & \mathbf{0} \end{bmatrix}$$

7 Process noise

$$\mathbf{q}[k] = \begin{bmatrix} \mathbf{0}^{\mathsf{T}} & \mathbf{0}^{\mathsf{T}} & -\boldsymbol{\varepsilon}_p^{\mathsf{T}} & \left(\hat{\bar{\mathbf{a}}}'\,\hat{\mathbf{w}}^{\mathsf{T}}\boldsymbol{\varepsilon}_p\right)^{\mathsf{T}} & \mathbf{w}_{a'}^{\mathsf{T}} \end{bmatrix}^{\mathsf{T}}$$

$$q_{a',j}[k] = 0 \qquad \text{(holding: } \bar{\mathbf{p}}_d \text{ fixed)}$$

$$q_{a',j}[k] = P_{\text{fix}}^2[k]\,I_{a',j}[k], \qquad j = 1,\ldots,6 \qquad \text{(regulation, scan)}$$

$$I_{a',j}[k] = \frac{1}{p_{a',j}[k]} - \frac{1}{p^f_{a',j}[k]}$$

$$\sqrt{P_{\text{fix}}}[k] = 0.4\left(1.1/h_d[k]\right)^2, \qquad h_d[k] = \hat{\mathbf{w}}^{\mathsf{T}}[k]\,\bar{\mathbf{p}}_d[k]$$

Observability, gates 1–2

$$\mathbf{F}_e(\mathbf{e}_a, \mathbf{e}_{a'}) = \left(\hat{\mathbf{w}}^{\mathsf{T}}\hat{\mathbf{M}}\right)\mathbf{I}_6, \qquad \mathbf{H}(\cdot, \mathbf{e}_{a'}) = \mathbf{0}$$

$$\mathbf{e}_a[k+N] - \mathbf{e}_a[k] \ni \mathbf{e}_{a'} \sum_{i=0}^{N-1} \hat{\mathbf{w}}^{\mathsf{T}}\hat{\mathbf{M}}[k+i]$$

gate 1: one path (state path), zero while $\hat{\mathbf{w}}^{\mathsf{T}}\hat{\mathbf{M}} = 0$ (hold, motion along the wall); gate 2: $\mathbf{e}_{a'}$ = ramp in $\mathbf{e}_a$ with the travel along the normal, $\mathbf{e}_a$ = constant offset; entry j of $\mathbf{e}_{a'}$ reaches only entry j of $\mathbf{e}_a$